

popquiz Newton's law I

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:33pm

Quiz Instructions

Can't help you here.

hint:

https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.html  (https://phet.colorado.edu/sims/html/forces-and-motion-basics/latest/forces-and-motion-basics_en.html)



Question 1 10 pts



This is a thought experiment done by Galielo. As discusses in class. So can't help you. There is no friction involved. (the corner is not supposed to be sharp). The ball rolls down and reach the flat surface. What happens next.

- ☐ the ball slows down and eventually stops since gravity stops pulling
- ☐ the ball keeping rolling in the same direction
- ☐ the ball stops abruptly
- ☐ can not be determined



I missed the class and didn't catch up



Question 2 17 pts



Let's say you try to push a large mass like an Anvil.

You are pushing forward (along the horizontal !!!) surface on Earth and then on the Moon. What is true?

So you are not lifting just pushing a mass forward. mass is inertia



Can not be determined



its easier on earth.



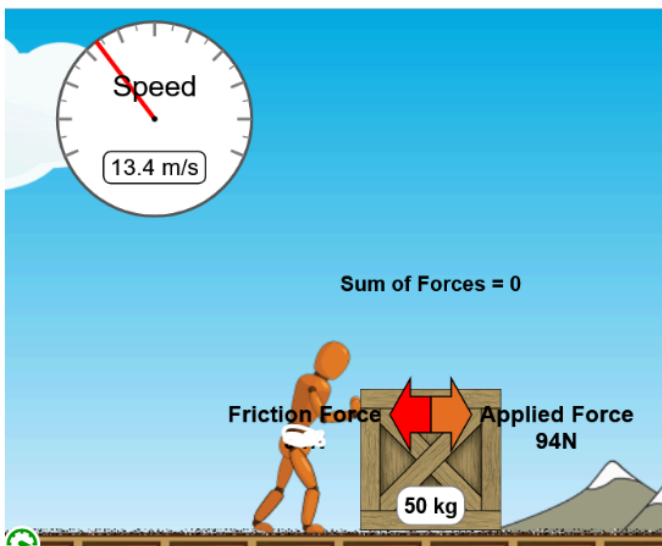
its easier on the Moon. gravity is divided by 6



It is as hard/ same mass



Question 3 17 pts



This crate is moving at a constant speed. What is the kinetic friction opposing the motion?

This crate is moving at a constant velocity. What is the kinetic friction opposing the motion?

hint: acceleration is 0 means constant velocity. So net force is 0



-94N



cant be determined



0N



188N



Question 4 10 pts

Building on the previous question. The person let go the box (so the only force acting is the kinetic friction). What is true? hint: $F_{\text{net}} = \text{mass} \times \text{acceleration}$. you have the force and the mass.

you can use the simulation (header)



The box keeps moving at the same speed

The box has an acceleration of 1.88 m/s/s 

The box stops moving instantaneously

The acceleration of the box will be -1.9 m/s/s 

Question 5 10 pts



As discussed in class. The airplane is taking off and is accelerating. The chairs applied a force on the passengers. What is true.

hint: $F = ma$. Same acceleration for both. Not the same mass.



they experience the same force



no. idea. I didn't catch up

- ☐ the force experienced by the kids is less than the force experienced by the adult.
- ☐ the force experienced by the adult is less than the force experienced by the kids



Question 6 10 pts

If the mass of an object is doubled while the force applied to it remains the same, what happens to its acceleration according to Newton's second law?

- A) The acceleration is halved
- B) The acceleration is quadrupled
- C) The acceleration remains the same
- D) The acceleration is doubled



C



D



A



B



Question 7 10 pts

A non-zero net force acts on an object. Does that mean the object necessarily moves in the same direction as the net force?

As discussed in class

hint: think projectile motion. in which direction is the weight? but in which direction the projectile is moving



Yes



I didn't catch up yet.



Whaaaaaaaaaaaaat



No



Question 8 10 pts

) The following four forces act on a 4.00 kg object:

$$\vec{F}_1 = 300 \text{ N east}$$

$$\vec{F}_2 = 700 \text{ N north}$$

$$\vec{F}_3 = 500 \text{ N west}$$

$$\vec{F}_4 = 600 \text{ N south}$$

What is the component of the net force. As discussed in class



(800, 1300) N



(-200, 100) N



(100, 200)



(400, 300) N



Question 9 10 pts

Building on the previous question. What is the coordinate of the acceleration



(25, 100) m/s/s



whaaaaaat?



(100, 75) m/s/s



(-50, 25) m/s/s



(200, 325) m/s/s

Not saved

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